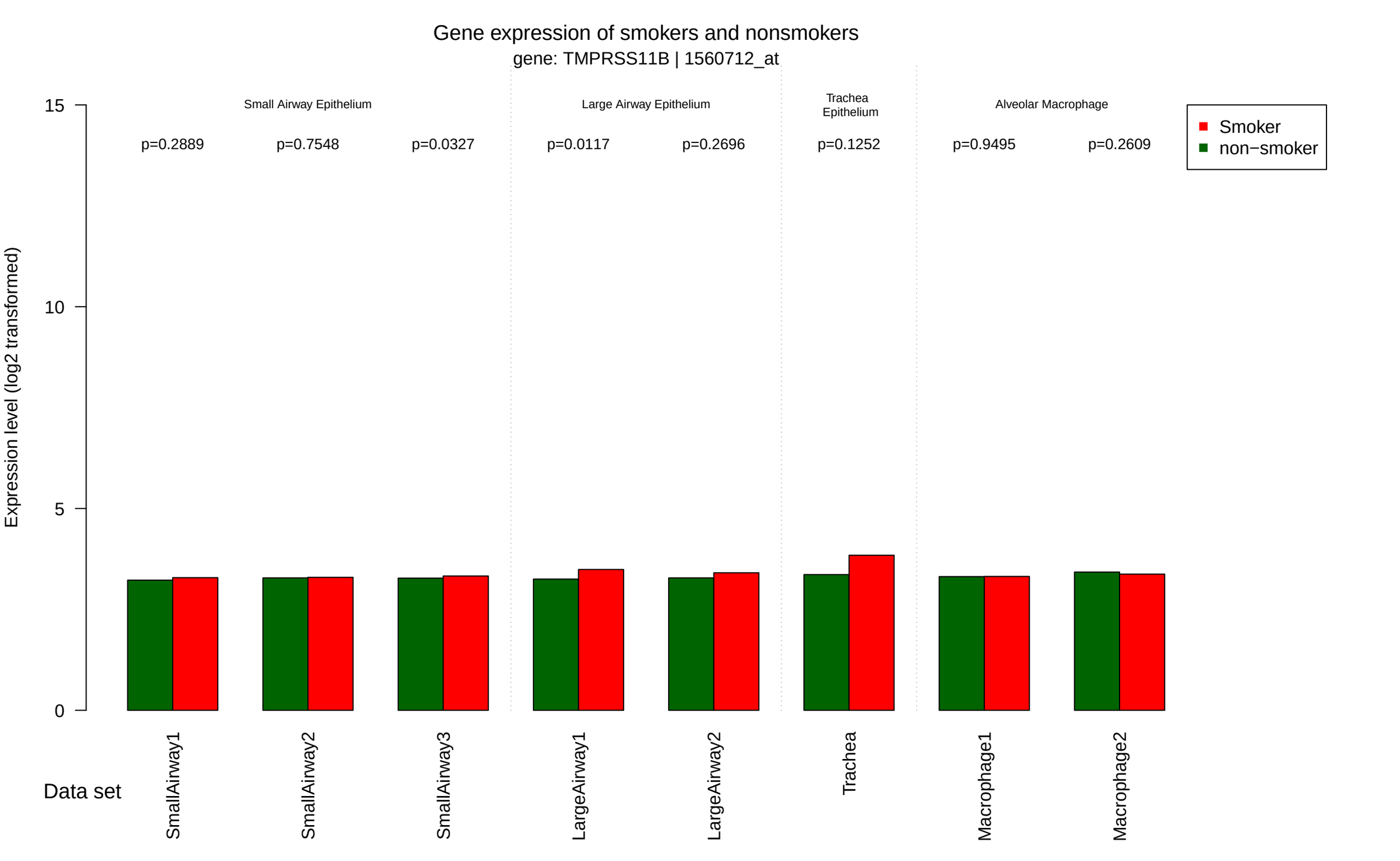
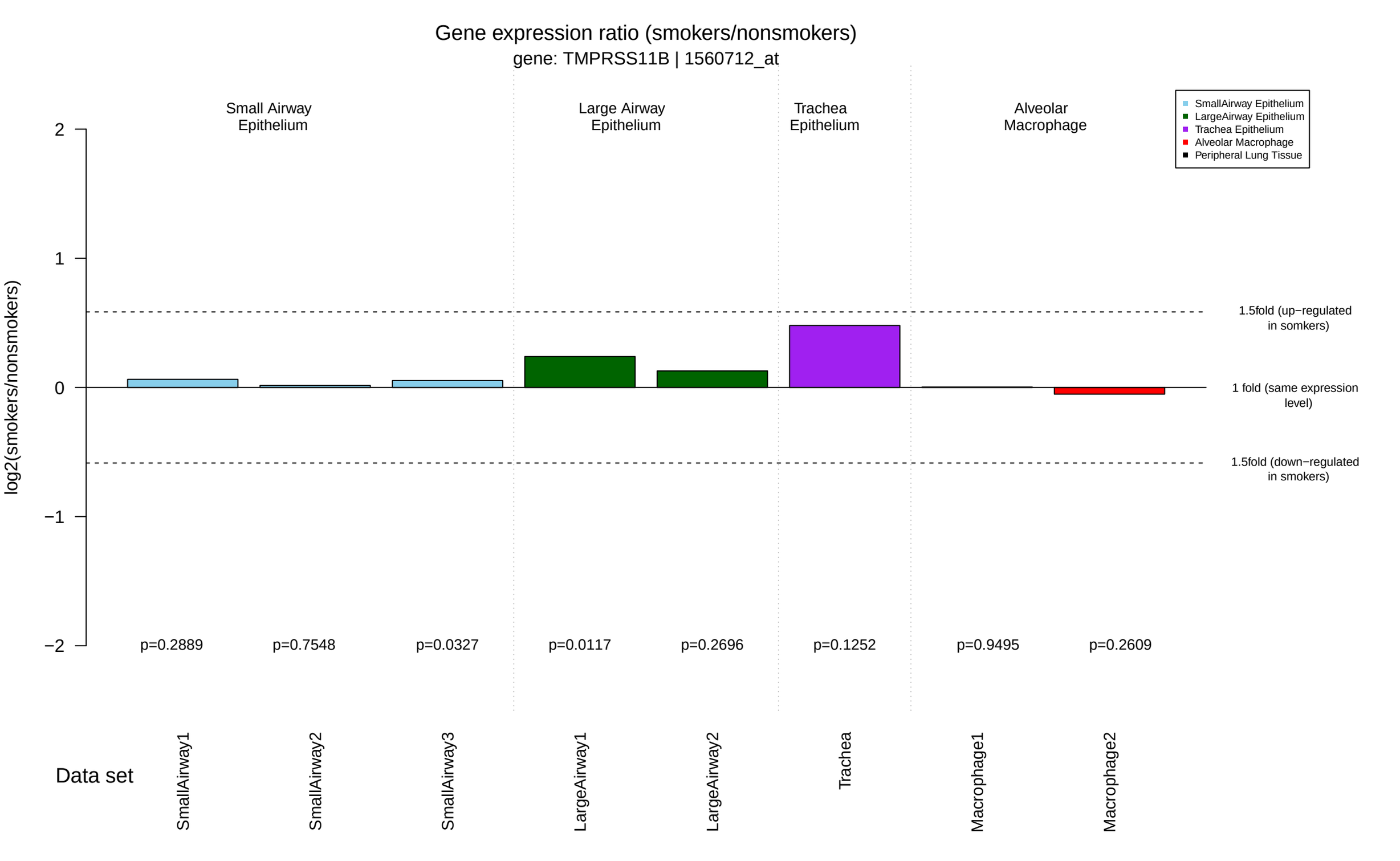
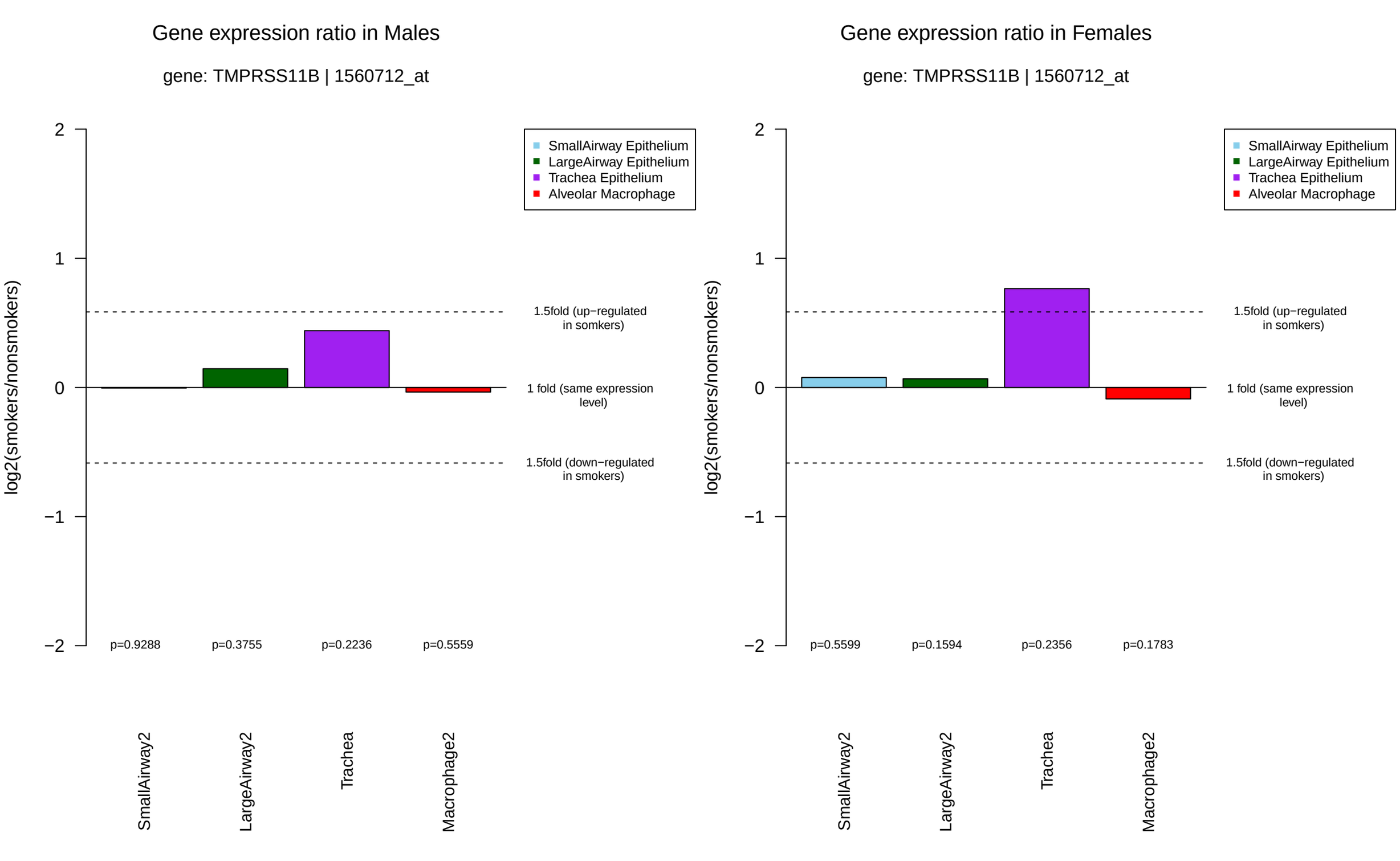
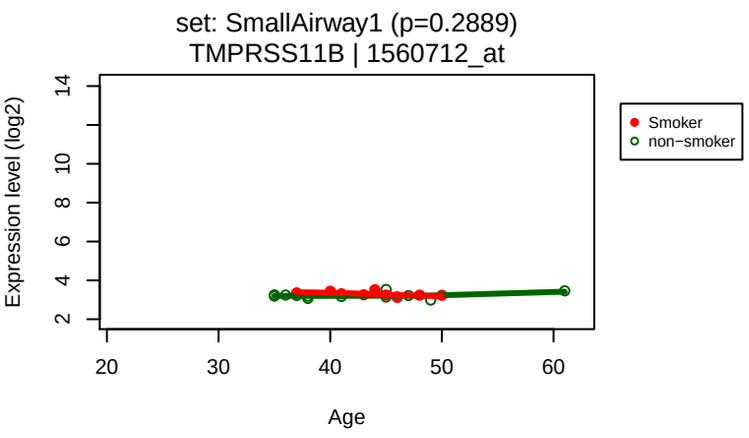
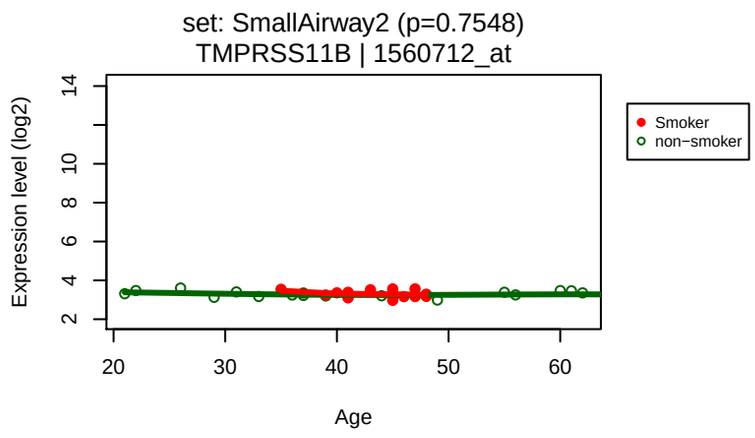
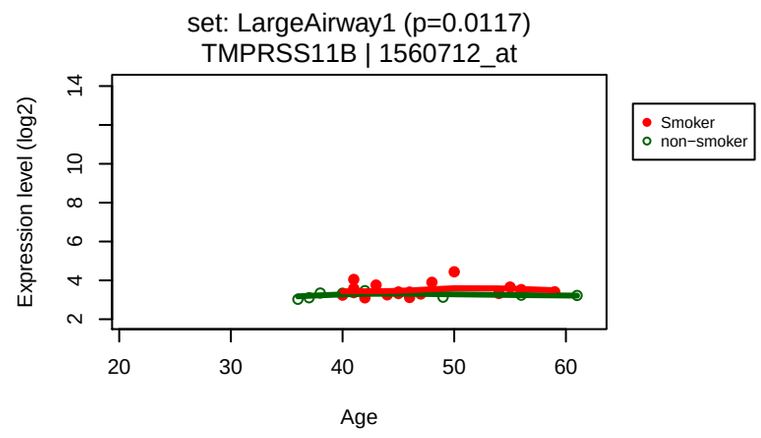

Gene expression of smokers and nonsmokers
gene: TMPRSS11B | 1560712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2889
p=0.7548
p=0.0327
p=0.0117
p=0.2696
p=0.1252
p=0.9495
p=0.2609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

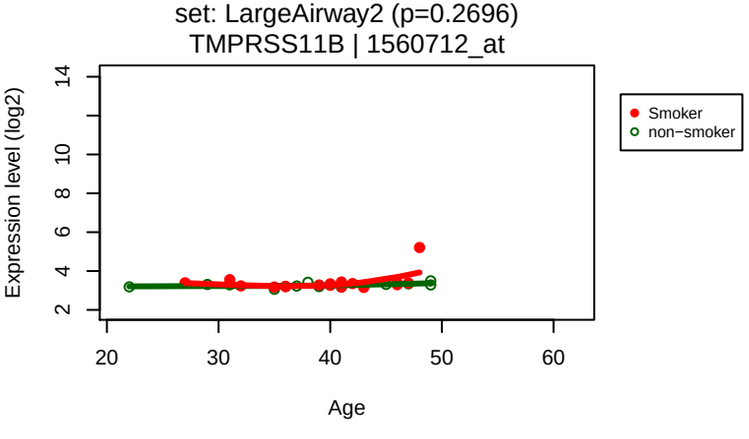

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS11B | 1560712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2889
p=0.7548
p=0.0327
p=0.0117
p=0.2696
p=0.1252
p=0.9495
p=0.2609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

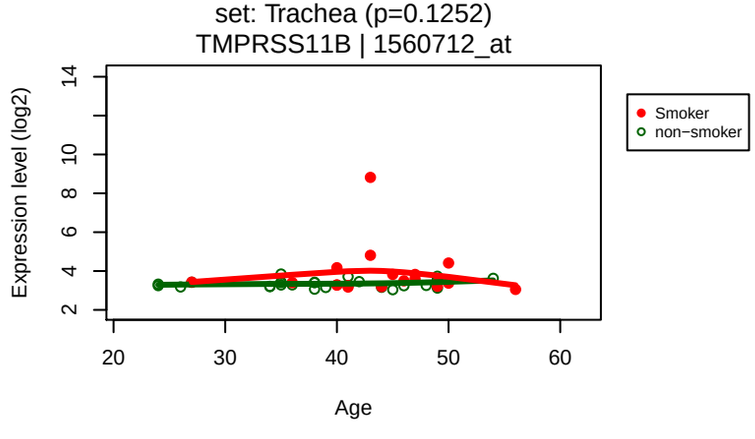

Gene expression ratio in Males
gene: TMPRSS11B | 1560712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9288
p=0.3755
p=0.2236
p=0.5559
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS11B | 1560712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5599
p=0.1594
p=0.2356
p=0.1783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

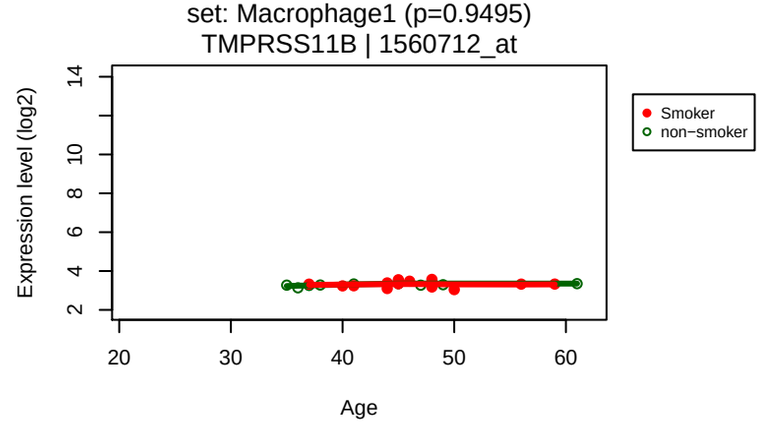

set: SmallAirway1 (p=0.2889)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker

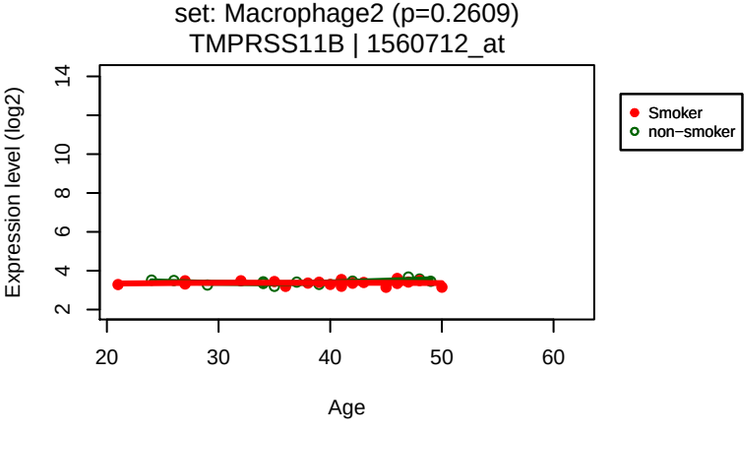

set: SmallAirway2 (p=0.7548)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker

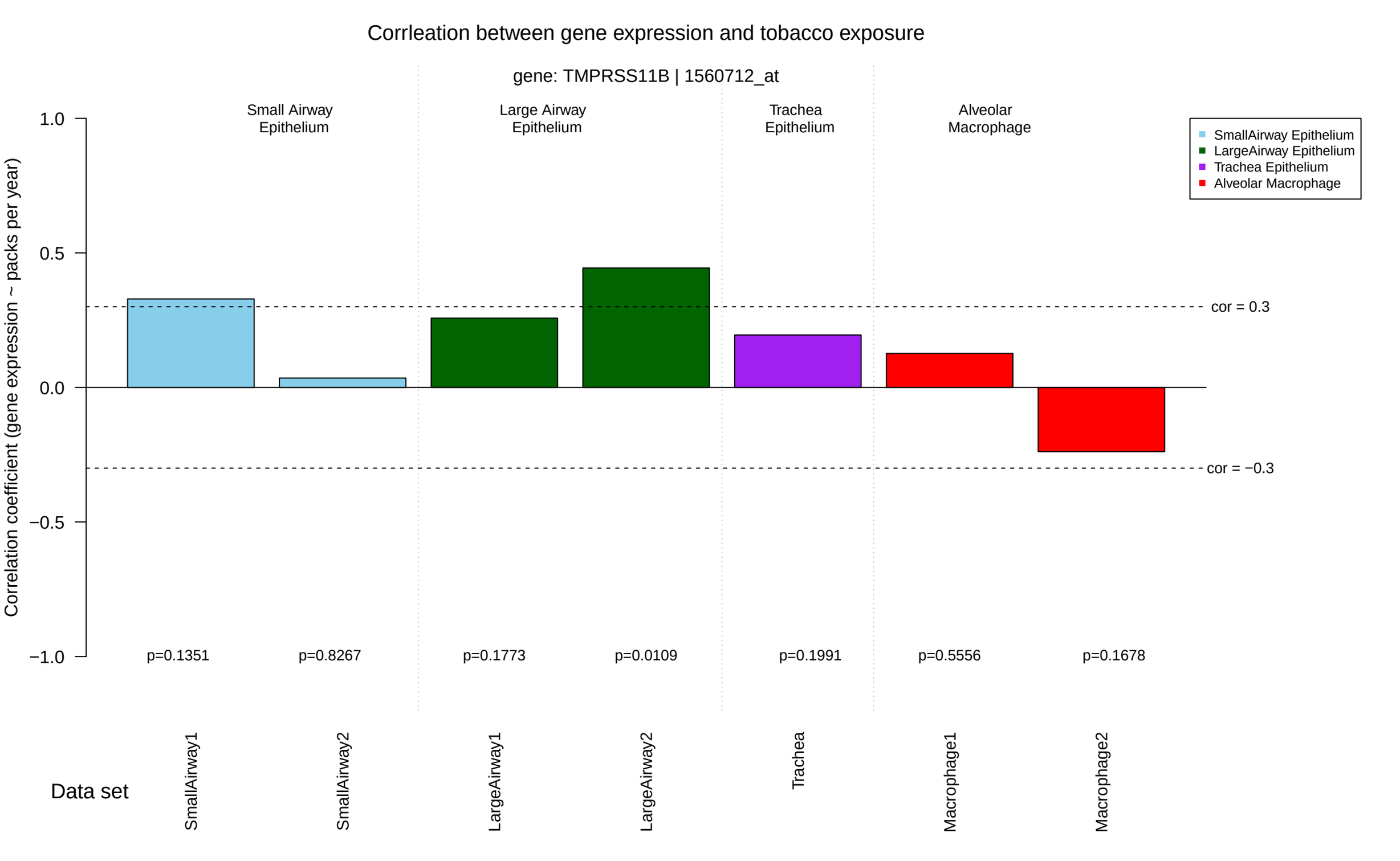

set: LargeAirway1 (p=0.0117)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2696)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1252)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9495)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2609)
TMPRSS11B | 1560712_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS11B | 1560712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1351
p=0.8267
p=0.1773
p=0.0109
p=0.1991
p=0.5556
p=0.1678
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2